

ANLP Assignment 1 Report

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1 Introduction

As part of the assignment, I had to implement a Transformer architecture from the paper [VSP⁺17] on a given corpus of Finnish English parallel sentences. Had to build a model for Finnish to English translation by using the Transformer architecture coded from scratch using pytorch. Had to experiment with two different Positional Embedding strategies and three different Decoding strategies. I only had to train the model for the two positional embeddings since the decoding strategies are used during inference time.

2 Architecture

The Transformer consists of Encoders and Decoders. The Encoder takes the source language tokens, embeds it into a rich embedding space by giving it complete bidirectional context, using multi headed attention. The Decoder considers the translated sentence so far and predicts the next word, autoregressively. It has a cross attention between the embeddings of the decoder input and all the words in the encoder. The K,V values used in the cross attention comes from

3 Results

- **RoPE clearly outperforms Relative Positional Encoding** across all decoding strategies:
 - Greedy: 3.10 vs. 1.78
 - Top-k: 2.48 vs. 1.68
 - Beam: 2.90 vs. 2.45
- **Greedy decoding gives the highest BLEU for RoPE** (3.10), while for Relative, **Beam search is best** (2.45).
- **Top-k sampling underperforms** for both encodings in BLEU (lowest scores). This is expected because top-k introduces randomness and tends to reduce n-gram overlap with reference translations.

Encoding	Greedy	Top-k	Beam
RoPE	3.10	2.48	2.90
Relative	1.78	1.68	2.45

Table 1: BLEU scores for RoPE and Relative positional encodings under different decoding strategies (higher is better).

References

- [VSP⁺17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

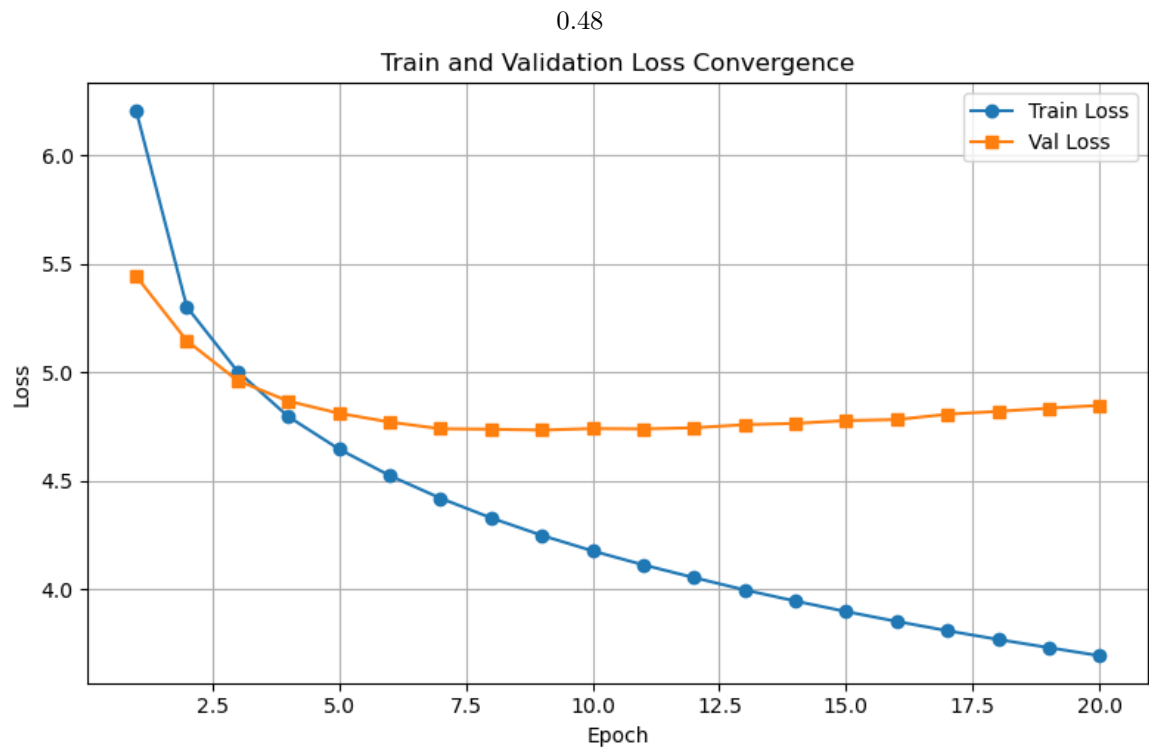


Figure 1: Loss convergence with RoPE.

0.48

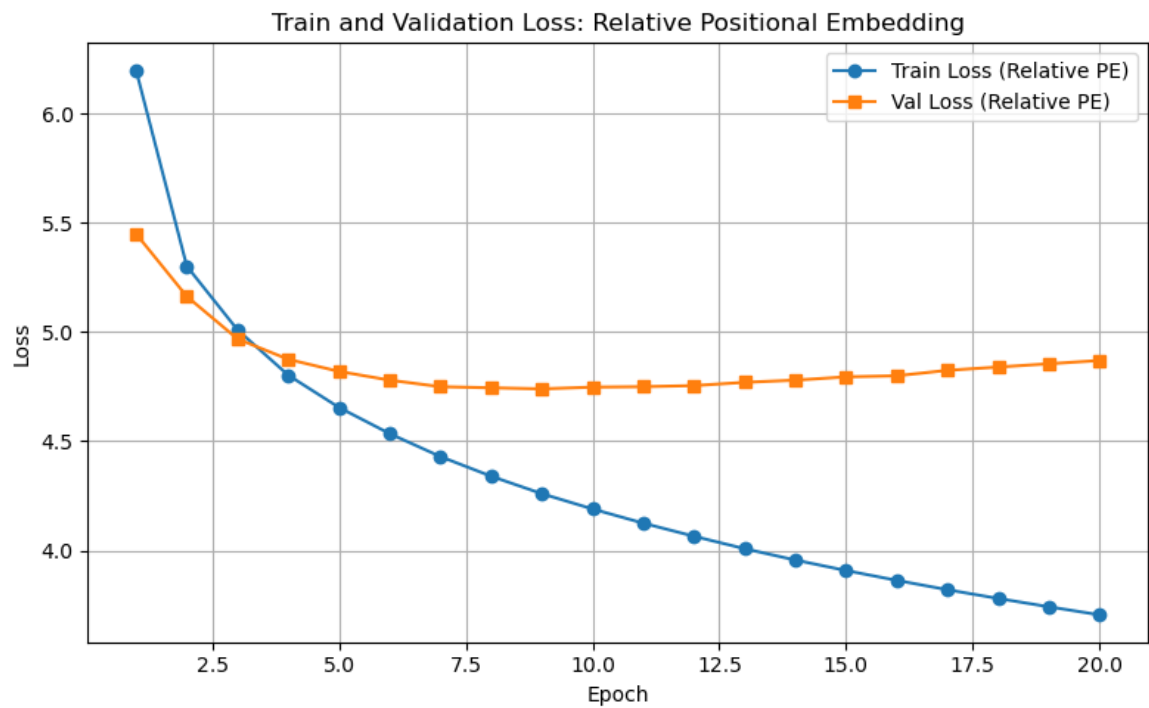


Figure 2: Loss convergence with Relative Positional Encoding.

Figure 3: Training loss convergence comparison between RoPE and Relative Positional Encoding.